Fuel System – General Data

1. General

Fuel specifications and tank capacities are given in Table 1 — Fuel Specifications and Grade and Table 2 — Fuel Tank Capacities and Fuel Weights JP–4 Fuel. The fuselage tank fuel filler caps are located on the left side of the fuselage, one forward of the wing strut and the other aft. When wing tanks are fitted (Mod S.O.O. 6247) a filler cap is located on each outer wing upper surface.

Table 1 Fuel Specifications and Grade

Canadian Specification	U.S. Specification	British Specification	NATO Symbol	Civil Specification
3–GP–22b	MIL–T–5624G	D Eng RD2486	F–40	Turbine Fuel JP–4
Note Aviation Kerosene or JP–5 may also be used. MIL–G–5572C AVGAS (all grades) is allowed for emergency use only and is limited to a total of 150 hours in any one overhaul cycle.				

Table 2 Fuel Tank Capacities and Fuel Weights JP–4 Fuel

Location	Approximate Weight (7.8 Lb./Imp Gal)	Capacity (Gallons) U.S.	Imperial
Forward Tank	1185	183	152
Aft Tank	1295	200	166
Wing Tanks (if fitted)	577	89	74
Total With Wing Tanks	3057	472	392
Total Without Wing Tanks	2480	383	318

Note Weight (approximately) of alternate fuels are as follows:

Aviation Kerosene	8.2 lb/Imperial Gallon
JP–5	8.3 lb/Imperial Gallon
AVGAS (Emergency)	7.2 lb/Imperial Gallon

2. Refueling

A. Refueling Precautions

WARNING A CHAMOIS OR FELT FILTER MUST NOT BE USED AS THESE CONTRIBUTE TO CONTAMINATION AND IN ADDITION, ELECTROSTATICALLY CHARGE THE FUEL.

Before attempting to refuel aircraft, observe the following precautions:

1) No smoking in the vicinity of the aircraft.

2) Position a suitable fire extinguisher in the immediate area of the aircraft, and ensure personnel involved are acquainted with its use.

3) Ground both aircraft and fuel tender.

4) Ensure aircraft electrical power is OFF.

5) Aircraft with Cold Weather Operations Kit installed (S.O.O. 6276 or 6287) ensure system is disconnected.

6) Before placing nozzle in the filler neck, ground nozzle.

7) Avoid carrying out other work on aircraft during refueling operations.

8) Avoid spilling fuel.

Note It is important that only clean fuel be used. If the presence of contaminants is suspected, tanks should be filled through a funnel with a 100 – 200 mesh metal screen.

B. Filler Neck Refueling

1) Observe refueling precautions. Refer to Para 2.A.

2) Ensure external power supply to aircraft is disconnected and aircraft supply is OFF.

3) Ground supply nozzle to ground socket, remove filler cap, insert nozzle and fill tank. When tank is full, remove nozzle, replace and secure filler cap and disconnect ground plug from ground socket.

4) Repeat Para 2.B.3) for other tank (and wing tanks if fitted) as necessary.

C. Alternate Method of Wing Tank Refueling

If over wing refueling of wing tanks cannot be carried out, the wing tanks may be refueled from the fuselage tanks as follows:

1) Apply DC power. (Refer to 12-00-00, Servicing – General Data, Para 2.).

2) Position fuel levers in the overhead console to OFF and select fuselage tank BOOST PUMP 1 switches to ON.

3) Wing fuel tank switches – REFUEL (left and/or right).

Note Using above procedure, the forward fuselage tank will refill the right wing tank and the rear fuselage tank will refill the left wing tank.

4) Observe that the wing tank fuel indications register gradual increase in contents, and that AFT and FWD (fuselage tank) fuel quantity indicators register a corresponding decrease. When tank content indicators show no further change, select wing fuel tank switch(es) and boost pump switch(es) to OFF.

Note As above operations require 15 to 20 minutes to completely fill a wing tank, an external power source should be used to power the electrical system.

3. Draining the Fuel System

CAUTION OBSERVE PRECAUTIONS DETAILED IN Para 2.A.

To drain fuselage fuel tanks, use the drains on the sump plates of fuel cells No. 4 and No. 5. No. 4 cell drains the forward tank and No. 5 cell drains the aft tank. To drain a wing tank (if fitted) switch on battery supply or connect an external electrical power source to aircraft, disconnect flexible hose at nacelle fuel filter tee (cap flexible hose), attach suitable length of hose to filter tee and run open end into container. Switch wing fuel tank switch to L or R ENGINE as applicable. When draining is complete select switch OFF, disconnect electrical supply, connect and secure flexible hose to filter tee.

4. Draining of Water and Contaminants from Fuel System

A. General

Drain valves are provided at various locations in the fuel system to allow removal of accumulated water and/or contaminants.

B. Locations of Water Drain Valves

Drain valves are located on the two main fuel filters and under No. 4 and No. 5 cells on the forward and aft fuselage tank interconnect manifolds, respectively, for all aircraft models.

On aircraft with Mod S.O.O. 6247 incorporated, an additional drain valve is provided on lower surface of each integral wing tank.

On aircraft with Mod S.O.O. 6048, an additional drain valve (Mod 6/1498) is provided under No. 8 cell on the aft fuselage tank interconnector manifold.

On aircraft with Mod 6/2209 incorporated, an additional drain valve is provided under No. 8 cell, on the forward and aft fuselage interconnector manifolds.

C. Draining Procedure

CAUTION OBSERVE PRECAUTIONS DETAILED IN Para 2.A.

Water and/or contaminants must be drained from integral wing tanks, fuel strainers, and fuselage tank interconnector manifolds at regular intervals as prescribed in the Inspection Schedule. To accomplish this, open water drain valves provided, keep valves open until clean fuel only is ejected and close valves.

Note Due to the higher locations of fuel filters in engine nacelles, water and/or contaminants from filters may be drawn into fuel tanks when associated drain valves are opened. To prevent contamination of fuel tanks, booster pumps must be operating before fuel filter drain valves are opened.

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